

The maps $\mathcal{L}_I \rightarrow \mathcal{L}_J$ here are induced by the ideal inclusions $\mathcal{O}_{\mathrm{FM}_n^{\mathrm{alg}}(X)}(-D(I)) \hookrightarrow \mathcal{O}_{\mathrm{FM}_n^{\mathrm{alg}}(X)}$. We call this screen the *universal screen* on $\mathrm{FM}_n^{\mathrm{alg}}(X)$.

Definition 4.2. Let $\mathcal{X}_n: \mathbf{Set} \rightarrow \mathbf{Sch}^{\mathrm{op}}$ be the functor which sends a scheme H to the set of screens on H up to isomorphism of the screen data.

Similarly let $\mathcal{X}(I_1, \dots, I_k)$ denote the subfunctor sending H to the set of screens on H up to isomorphism which satisfy the I_r -vanishing property for each $1 \leq r \leq k$.

Proposition 4.3. *The Fulton-MacPherson compactification $\mathrm{FM}_n^{\mathrm{alg}}(X)$ represents the functor $\mathcal{X}_n$. Furthermore, the natural isomorphism $\mathrm{hom}(-, \mathrm{FM}_n^{\mathrm{alg}}(X)) \rightarrow \mathcal{X}_n$ is given by sending the element $f: H \rightarrow \mathrm{FM}_n^{\mathrm{alg}}(X)$ to the pullback to H of the universal screen on $\mathrm{FM}_n^{\mathrm{alg}}(X)$ described above.*

Similarly $D(I_1, \dots, I_n) \subseteq \mathrm{FM}_n^{\mathrm{alg}}(X)$ represents $\mathcal{X}(I_1, \dots, I_k)$. The natural isomorphism is given in the same way here as above.

Proof. This is theorem 4 in [FM94]. □

4.2 Definition and Properties of $T_{d,n}$

In this section we recall some properties of the $T_{d,n}$ spaces introduced by Chen, Gibney, and Krashen [CGK09]. We also introduce a functor of points for these spaces and use this to define an operad with spaces $\{T_{d,n}\}_{n \in \mathbb{N}}$.

If $\mathrm{pr}_i: X^n \rightarrow X$ denotes the projection to the i :th component it is clear from definitions that the composition

$$D([n]) \hookrightarrow \mathrm{FM}_n^{\mathrm{alg}}(X) \xrightarrow{\rho} X^n \xrightarrow{\mathrm{pr}_i} X$$

is independent of the integer i . We denote this morphism by $\pi: D([n]) \rightarrow X$. In [CGK09], Chen, Gibney, and Krashen show that for a fixed n , any smooth scheme X of dimension d , and any point $x \in X$, the fibers $\pi^{-1}(x)$ are isomorphic. We define $T_{d,n}$ as $\pi^{-1}(0)$ for the map $\pi: D([n]) \rightarrow \mathbb{A}^d$ but by the above we could equally well have taken any other smooth variety instead of $X = \mathbb{A}^d$.

Remark. Just like the topological case notice that while we have up until this point indexed coordinates with $[n]$ we are free to use any finite index set I and $\mathrm{FM}_I^{\mathrm{alg}}(X)$ and $T_{d,I}$ together with all related maps and divisors are defined in the analogous way.

The variety $T_{d,n}$ is smooth and comes equipped with a smooth closed subscheme $T_{d,n}(I)$ for every non empty $I \subseteq [n]$. For $I \in P_2(n)$, i.e. $I \neq [n]$ and $|I| \geq 2$, we define $T_{d,n}(I)$ as the pullback of $D(I) \subseteq \mathrm{FM}_n^{\mathrm{alg}}(X)$ to $T_{d,n}$. For other I (i.e. I is $[n]$ or a one point set) we take $T_{d,n}(I) := T_{d,n}$. Similarly, for a collection of (distinct) subsets $I_1, \dots, I_k \subseteq [n]$ we define $T_{d,n}(I_1, \dots, I_k) := \bigcap_i T_{d,n}(I_i)$. For $I \neq [n]$ and $|I| \geq 2$ the subscheme $T_{d,n}(I)$ is a smooth divisor of $T_{d,n}$ and the union of all these divisors is a smooth normal crossings divisor. Additionally, the intersection $T_{d,n}(I_1, \dots, I_k)$ is non empty if and only if $(I_1, \dots, I_k)$ is *nested*. All this essentially follows from the analogous results for $\mathrm{FM}_n^{\mathrm{alg}}(X)$ recalled in Section 4.1. Details can be found in [CGK09].

Remark. The reason we include definitions of $T_{d,n}(\{i\})$ and $T_{d,n}([n])$ is that this will simplify the notation for some later results.

The first goal of this section will be to give a functor of points description of the schemes $T_{d,n}(I_1, \dots, I_k)$. For this I will introduce the following

Definition 4.4. For a scheme H , an integer d , and a finite, non-empty, set I with at least two element we let $\mathcal{F}_I^{H,d}$ denote the quasi-coherent sheaf of modules defined by

$$\mathcal{F}_I^{H,d} := \left\langle \{t_{ij}^k\}_{i,j \in I}^{1 \leq k \leq d} \right\rangle / \sim.$$

Here $\left\langle \{t_{ij}^k\}_{i,j \in I}^{1 \leq k \leq d} \right\rangle$ denotes the free module with generators t_{ij}^k and $\sim$ is generated by sections of the form $t_{ij}^k + t_{jl}^k - t_{il}^k$. We will also denote this by $\mathcal{F}_I^d$ or $\mathcal{F}_I$ if H or H, d are clear from context.